

lab_code_21.c

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1 // Name: Md Shagor
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4 // Lab Report: 21
5 // Experiment Name: Swapping two variable values using pointers
6
7
8 #include <stdio.h>
9
10 /**
11  * Experiment: Swapping two numbers using Call by Reference
12  * Logic: Use pointers to access and modify the actual memory addresses of variables.
13  */
14
15 // Function to swap values using pointers
16 void swap(int *x, int *y) {
17     int temp;
18     temp = *x; // temp holds the value at address x
19     *x = *y;   // value at address x is replaced by value at address y
20     *y = temp; // value at address y is replaced by temp
21 }
22
23 int main() {
24     int a, b;
25
26     // 1. User Input
27     printf("Enter the value for A: ");
28     if (scanf("%d", &a) != 1) return 1;
29
30     printf("Enter the value for B: ");
31     if (scanf("%d", &b) != 1) return 1;
32
33     // 2. Display Before Swap
34     printf("\n--- Before Swapping ---\n");
35     printf("A = %d, B = %d\n", a, b);
36
37     // 3. Perform Swap
38     // We pass the ADDRESSES (&) so the function can reach into main's memory
39     swap(&a, &b);
40
41     // 4. Display After Swap
42     printf("\n--- After Swapping ---\n");
43     printf("A = %d, B = %d\n", a, b);
44
45     return 0;
46 }
```